

SAFETY DATA SHEET



MARSHAL® 250 EC

Version	Revision Date:	SDS Number:	Date of last issue: -
1.0	24.09.2025	50000504	Date of first issue: 24.09.2025

SECTION 1: Identification of the substance/mixture and of the company/undertaking

1.1 Product identifier

Product name MARSHAL® 250 EC

Other means of identification

Product code 50000504

1.2 Relevant identified uses of the substance or mixture and uses advised against

Use of the Sub- : Insecticide
stance/Mixture

Recommended restrictions : Use as recommended by the label.
on use

1.3 Details of the supplier of the safety data sheet

Supplier Address FMC Agricultural Solutions A/S
Thyborønvej 78
DK-7673 Harbøre
Denmark

Telephone: +45 9690 9690
Telefax: +45 9690 9691
E-mail address: SDS-Info@fmc.com .

1.4 Emergency telephone

For leak, fire, spill or accident emergencies, call:
1 703 / 741-5970 (CHEMTREC - International)
1 703 / 527-3887 (CHEMTREC - Alternate)

Medical emergency:
In case of poisoning call toll free number: 0800720021 or
0800730030 (24 hrs/masaa 24)

SECTION 2: Hazards identification

2.1 Classification of the substance or mixture

Classification (REGULATION (EC) No 1272/2008)

Flammable liquids, Category 3 H226: Flammable liquid and vapor.

Acute toxicity, Category 3 H301: Toxic if swallowed.

SAFETY DATA SHEET



MARSHAL® 250 EC

Version 1.0	Revision Date: 24.09.2025	SDS Number: 50000504	Date of last issue: - Date of first issue: 24.09.2025
----------------	------------------------------	-------------------------	--

Acute toxicity, Category 2	H330: Fatal if inhaled.
Acute toxicity, Category 4	H312: Harmful in contact with skin.
Eye irritation, Category 2	H319: Causes serious eye irritation.
Skin sensitization, Sub-category 1B	H317: May cause an allergic skin reaction.
Carcinogenicity, Category 2	H351: Suspected of causing cancer.
Specific target organ toxicity - single exposure, Category 1	H370: Causes damage to organs.
Specific target organ toxicity - single exposure, Category 3, Respiratory system	H335: May cause respiratory irritation.
Specific target organ toxicity - single exposure, Category 3, Central nervous system	H336: May cause drowsiness or dizziness.
Specific target organ toxicity - repeated exposure, Category 1	H372: Causes damage to organs through prolonged or repeated exposure.
Aspiration hazard, Category 1	H304: May be fatal if swallowed and enters airways.
Short-term (acute) aquatic hazard, Category 1	H400: Very toxic to aquatic life.
Long-term (chronic) aquatic hazard, Category 1	H410: Very toxic to aquatic life with long lasting effects.

2.2 Label elements

Labeling (REGULATION (EC) No 1272/2008)

Hazard pictograms :



Signal Word : DANGER

Hazard Statements :

H226	Flammable liquid and vapor.
H301	Toxic if swallowed.
H304	May be fatal if swallowed and enters airways.
H312	Harmful in contact with skin.
H317	May cause an allergic skin reaction.
H319	Causes serious eye irritation.
H330	Fatal if inhaled.
H335	May cause respiratory irritation.
H336	May cause drowsiness or dizziness.
H351	Suspected of causing cancer.
H370	Causes damage to organs.

SAFETY DATA SHEET



MARSHAL® 250 EC

Version	Revision Date:	SDS Number:	Date of last issue: -
1.0	24.09.2025	50000504	Date of first issue: 24.09.2025

H372 Causes damage to organs through prolonged or repeated exposure.

H410 Very toxic to aquatic life with long lasting effects.

Precautionary Statements :

Prevention:

P210 Keep away from heat, hot surfaces, sparks, open flames and other ignition sources. No smoking.

P260 Do not breathe mist or vapors.

P273 Avoid release to the environment.

P280 Wear protective gloves/ protective clothing/ eye protection/ face protection/ hearing protection.

Response:

P301 + P310 + P330 IF SWALLOWED: Immediately call a POISON CENTER/ doctor. Rinse mouth.

P304 + P340 + P310 IF INHALED: Remove person to fresh air and keep comfortable for breathing. Immediately call a POISON CENTER/ doctor.

P308 + P311 IF exposed or concerned: Call a POISON CENTER/ doctor.

P331 Do NOT induce vomiting.

P370 + P378 In case of fire: Use dry sand, dry chemical or alcohol-resistant foam to extinguish.

P391 Collect spillage.

Storage:

P403 + P233 Store in a well-ventilated place. Keep container tightly closed.

Hazardous ingredients which must be listed on the label:

Aromatic hydrocarbons, C9; Alkylbenzenes; C9-aromatics

Distillates (petroleum), solvent-dewaxed light paraffinic; Baseoil — unspecified carbosulfan (ISO)

Benzenesulfonic acid, mono-C11-13-branched alkyl derivs., calcium salts

Additional Labeling

Restricted to professional users.

EUH066 Repeated exposure may cause skin dryness or cracking.

EUH401 To avoid risks to human health and the environment, comply with the instructions for use.

2.3 Other hazards

This substance/mixture contains no components considered to be either persistent, bioaccumulative and toxic (PBT), or very persistent and very bioaccumulative (vPvB) at levels of 0.1% or higher.

SAFETY DATA SHEET



MARSHAL® 250 EC

Version 1.0 Revision Date: 24.09.2025 SDS Number: 50000504 Date of last issue: -
Date of first issue: 24.09.2025

SECTION 3: Composition/information on ingredients

3.2 Mixtures

Components

Chemical name	CAS-No. EC-No. Index-No. Registration number	Classification	Concentration (% w/w)
Aromatic hydrocarbons, C9; Alkylbenzenes; C9-aromatics	128601-23-0	Flam. Liq. 3; H226 STOT SE 3; H335 (Respiratory system) STOT SE 3; H336 (Central nervous system) Asp. Tox. 1; H304 Aquatic Chronic 2; H411	>= 30 - < 50
Distillates (petroleum), solvent-dewaxed light paraffinic; Baseoil — unspecified	64742-56-9 265-159-2 649-469-00-9	Carc. 1B; H350 Asp. Tox. 1; H304	>= 20 - < 30
carbosulfan (ISO)	55285-14-8 259-565-9 006-084-00-5	Acute Tox. 3; H301 Acute Tox. 2; H330 Skin Sens. 1; H317 STOT SE 1; H370 (Nervous system, Bladder, Gastro-intestinal system, Blood) STOT RE 1; H372 (Nervous system, Bladder, Gastro-intestinal system, Blood) Aquatic Acute 1; H400 Aquatic Chronic 1; H410 M-Factor (Acute aquatic toxicity): 100 M-Factor (Chronic aquatic toxicity): 10	>= 25 - < 30
Sorbitan, tri-(9Z)-9-octadecenoate, poly(oxy-1,2-ethanediyl) derivs.	9005-70-3	Aquatic Chronic 3; H412	>= 1 - < 2.5
Benzenesulfonic acid, mono-C11-13-branched alkyl derivs., calcium salts	68953-96-8 273-234-6	Acute Tox. 4; H312 Skin Irrit. 2; H315 Eye Dam. 1; H318 Aquatic Chronic 2;	>= 1 - < 2.5

SAFETY DATA SHEET



MARSHAL® 250 EC

Version	Revision Date:	SDS Number:	Date of last issue: -
1.0	24.09.2025	50000504	Date of first issue: 24.09.2025

		H411	
--	--	------	--

For explanation of abbreviations see section 16.

SECTION 4: First aid measures

4.1 Description of first-aid measures

- | | |
|----------------------------|--|
| General advice | : Move out of dangerous area.
Consult a physician.
Show this material safety data sheet to the doctor in attendance.
Symptoms of poisoning may appear several hours later.
Do not leave the victim unattended. |
| Protection of first-aiders | : Avoid inhalation, ingestion and contact with skin and eyes. |
| If inhaled | : Remove to fresh air.
Call a physician or poison control center immediately.
If unconscious, place in recovery position and seek medical advice. |
| In case of skin contact | : Take victim immediately to hospital.
If on skin, rinse well with water.
If on clothes, remove clothes.
Wash contaminated clothing before re-use. |
| In case of eye contact | : Immediately flush eye(s) with plenty of water.
Remove contact lenses.
Protect unharmed eye.
Keep eye wide open while rinsing.
If eye irritation persists, consult a specialist. |
| If swallowed | : Clean mouth with water and drink afterwards plenty of water.
Keep respiratory tract clear.
Do NOT induce vomiting.
Do not give milk or alcoholic beverages.
Never give anything by mouth to an unconscious person.
Take victim immediately to hospital. |

4.2 Most important symptoms and effects, both acute and delayed

- | | |
|----------|--|
| Symptoms | : Symptoms of overexposure include headache, light-headedness, weakness, abdominal cramps, nausea, excessive salivation, perspiration, blurred vision, tearing, pin-point pupils, blue skin color, convulsions, tremor and coma. |
| Risks | : The product contains petroleum distillates, which may pose an aspiration pneumonia hazard.

Toxic if swallowed.
May be fatal if swallowed and enters airways. |

MARSHAL® 250 EC

Version	Revision Date:	SDS Number:	Date of last issue: -
1.0	24.09.2025	50000504	Date of first issue: 24.09.2025

Harmful in contact with skin.
May cause an allergic skin reaction.
Causes serious eye irritation.
Fatal if inhaled.
May cause respiratory irritation.
May cause drowsiness or dizziness.
Suspected of causing cancer.
Causes damage to organs.
Causes damage to organs through prolonged or repeated exposure.
Repeated exposure may cause skin dryness or cracking.

4.3 Indication of any immediate medical attention and special treatment needed

Treatment : Treat symptomatically.
This product contains a reversible cholinesterase inhibitor. Atropine sulfate is antidotal. If cyanosis is absent: Adults - start treatment by giving 2 mg atropine intravenously or intramuscularly, if necessary, and repeat with 0.4 - 2.0 mg atropine at 15 minute intervals until atropinization occurs (tachycardia, flushed skin, dry mouth, mydriasis); Children under 12 - initial dose = 0.05 mg/kg body weight and repeat dose = 0.02 - 0.05 mg/kg body weight. Use of oximes such as 2-PAM is controversial. Observe patient to ensure that these symptoms do not recur as atropinization wears off. If in eyes, instill one drop of homatropine. Treatment is otherwise controlled removal of exposure followed by symptomatic and supportive care.

SECTION 5: Firefighting measures**5.1 Extinguishing media**

Suitable extinguishing media : Alcohol-resistant foam
Carbon dioxide (CO₂)
Dry chemical
Use extinguishing measures that are appropriate to local circumstances and the surrounding environment.

Unsuitable extinguishing media : Do not spread spilled material with high-pressure water streams.
High volume water jet

5.2 Special hazards arising from the substance or mixture

Specific hazards during fire fighting : Do not allow run-off from fire fighting to enter drains or water courses.

Hazardous combustion products : Fire may produce irritating, corrosive and/or toxic gases.
Carbon oxides
Sulfur oxides
Nitrogen oxides (NO_x)
Hydrogen cyanide

MARSHAL® 250 EC

Version	Revision Date:	SDS Number:	Date of last issue: -
1.0	24.09.2025	50000504	Date of first issue: 24.09.2025

5.3 Advice for firefighters

- Special protective equipment for fire-fighters : Firefighters should wear protective clothing and self-contained breathing apparatus.
- Further information : Collect contaminated fire extinguishing water separately. This must not be discharged into drains.
Fire residues and contaminated fire extinguishing water must be disposed of in accordance with local regulations.
For safety reasons in case of fire, cans should be stored separately in closed containments.
Use a water spray to cool fully closed containers.
-

SECTION 6: Accidental release measures**6.1 Personal precautions, protective equipment and emergency procedures**

- Personal precautions : Use personal protective equipment.
Ensure adequate ventilation.
Remove all sources of ignition.
Evacuate personnel to safe areas.
Beware of vapors accumulating to form explosive concentrations. Vapors can accumulate in low areas.
Immediately evacuate personnel to safe areas.
Never return spills in original containers for re-use.
Mark the contaminated area with signs and prevent access to unauthorized personnel.
Only qualified personnel equipped with suitable protective equipment may intervene.

6.2 Environmental precautions

- Environmental precautions : Prevent product from entering drains.
Prevent further leakage or spillage if safe to do so.
If the product contaminates rivers and lakes or drains inform respective authorities.

6.3 Methods and material for containment and cleaning up

- Methods for cleaning up : Contain spillage, and then collect with non-combustible absorbent material, (e.g. sand, earth, diatomaceous earth, vermiculite) and place in container for disposal according to local / national regulations (see section 13).

6.4 Reference to other sections

See sections: 7, 8, 11, 12 and 13.

SECTION 7: Handling and storage**7.1 Precautions for safe handling**

- Advice on safe handling : Avoid formation of aerosol.
Do not breathe vapors/dust.
-

MARSHAL® 250 EC

Version	Revision Date:	SDS Number:	Date of last issue: -
1.0	24.09.2025	50000504	Date of first issue: 24.09.2025

Avoid exposure - obtain special instructions before use.
 Avoid contact with skin and eyes.
 For personal protection see section 8.
 Smoking, eating and drinking should be prohibited in the application area.
 Take precautionary measures against static discharges.
 Provide sufficient air exchange and/or exhaust in work rooms.
 Open drum carefully as content may be under pressure.
 Dispose of rinse water in accordance with local and national regulations.
 Persons susceptible to skin sensitization problems or asthma, allergies, chronic or recurrent respiratory disease should not be employed in any process in which this mixture is being used.

Advice on protection against fire and explosion : Do not spray on a naked flame or any incandescent material. Take necessary action to avoid static electricity discharge (which might cause ignition of organic vapors). Keep away from open flames, hot surfaces and sources of ignition.

Hygiene measures : Avoid contact with skin, eyes and clothing. When using do not eat or drink. When using do not smoke. Wash hands before breaks and immediately after handling the product.

7.2 Conditions for safe storage, including any incompatibilities

Requirements for storage areas and containers : Prevent unauthorized access. No smoking. Keep container tightly closed in a dry and well-ventilated place. Containers which are opened must be carefully resealed and kept upright to prevent leakage. Observe label precautions. Electrical installations / working materials must comply with the technological safety standards.

Further information on storage conditions : Store in closed, labelled containers. The storage room should be constructed of incombustible material, closed, dry, ventilated and with impermeable floor, without access of unauthorised persons or children. A warning sign reading "POISON" is recommended. The room should only be used for storage of chemicals. Food, drink, feed and seed should not be present. A hand wash station should be available.

Further information on storage stability : No decomposition if stored and applied as directed.

7.3 Specific end use(s)

Specific use(s) : Registered pesticide to be used in accordance with a label approved by country-specific regulatory authorities.

Version 1.0 Revision Date: 24.09.2025 SDS Number: 50000504 Date of last issue: -
Date of first issue: 24.09.2025

SECTION 8: Exposure controls/personal protection

8.1 Control parameters

Occupational Exposure Limits

Contains no substances with occupational exposure limit values.

Derived No Effect Level (DNEL) according to Regulation (EC) No. 1907/2006

Substance name	End Use	Routes of exposure	Potential health effects	Value
Aromatic hydrocarbons, C9; Alkylbenzenes; C9-aromatics	Workers	Inhalation	Long-term systemic effects	150 mg/m3
	Workers	Dermal	Long-term systemic effects	25 mg/kg bw/day
	Consumers	Inhalation	Long-term systemic effects	32 mg/m3
	Consumers	Dermal	Long-term systemic effects	11 mg/kg bw/day
	Consumers	Oral	Long-term systemic effects	11 mg/kg bw/day
Distillates (petroleum), solvent-dewaxed light paraffinic; Baseoil — unspecified	Workers	Inhalation		2.7 mg/m3
	Workers	Dermal		1 mg/kg
	Consumers	Oral		0.74 mg/kg
Soybean oil, epoxidized	Workers	Inhalation	Long-term systemic effects	11.9 mg/m3
	Workers	Inhalation	Acute systemic effects	70 mg/m3
	Workers	Dermal	Long-term systemic effects	1.7 mg/kg bw/day
	Workers	Dermal	Acute systemic effects	10 mg/kg bw/day
	Consumers	Inhalation	Long-term systemic effects	2.8 mg/m3
	Consumers	Inhalation	Acute systemic effects	17.5 mg/m3
	Consumers	Dermal	Long-term systemic effects	0.800 mg/kg bw/day
	Consumers	Dermal	Acute systemic effects	5 mg/kg bw/day
	Consumers	Oral	Long-term systemic effects	0.800 mg/kg bw/day
	Consumers	Oral	Acute systemic effects	5 mg/kg bw/day
Benzenesulfonic acid, mono-C11-13-branched alkyl derivs., calcium salts	Workers	Inhalation	Long-term systemic effects	6 mg/m3
	Workers	Dermal	Long-term systemic	8.5 mg/kg

SAFETY DATA SHEET



MARSHAL® 250 EC

Version 1.0 Revision Date: 24.09.2025 SDS Number: 50000504 Date of last issue: -
Date of first issue: 24.09.2025

			effects	bw/day
	Consumers	Inhalation	Long-term systemic effects	1.48 mg/m3
	Consumers	Dermal	Long-term systemic effects	4.25 mg/kg bw/day
	Consumers	Oral	Long-term systemic effects	0.43 mg/kg bw/day

Predicted No Effect Concentration (PNEC) according to Regulation (EC) No. 1907/2006

Substance name	Environmental Compartment	Value
Distillates (petroleum), solvent-dewaxed light paraffinic; Baseoil — unspecified	Oral	9.33 mg/kg
Soybean oil, epoxidized	Soil	6.25 mg/kg dry weight (d.w.)
Benzenesulfonic acid, mono-C11-13-branched alkyl derivs., calcium salts	Fresh water	0.023 mg/l
	Sea water	0.002 mg/l
	Sewage treatment plant	5.5 mg/l
	Fresh water sediment	1.35 mg/kg
	Sea sediment	0.135 mg/kg
	Soil	0.124 mg/kg
	Intermittent use (freshwater)	0.290 mg/l

8.2 Exposure controls**Personal protective equipment**

- Eye/face protection : Eye wash bottle with pure water
Tightly fitting safety goggles
Wear face-shield and protective suit for abnormal processing problems.
- Hand protection
Material : Wear chemical resistant gloves, such as barrier laminate, butyl rubber or nitrile rubber.
- Remarks : The suitability for a specific workplace should be discussed with the producers of the protective gloves.
- Skin and body protection : Impervious clothing
Choose body protection according to the amount and concentration of the dangerous substance at the work place.
- Respiratory protection : In the case of dust or aerosol formation use respirator with an approved filter.
- Protective measures : Plan first aid action before beginning work with this product.
Always have on hand a first-aid kit, together with proper instructions.
Wear suitable protective equipment.
When using do not eat, drink or smoke.
In the context of professional plant protection use as recommended, the end user must refer to the label and the instruc-

SAFETY DATA SHEET



MARSHAL® 250 EC

Version	Revision Date:	SDS Number:	Date of last issue: -
1.0	24.09.2025	50000504	Date of first issue: 24.09.2025

tions for use.

SECTION 9: Physical and chemical properties

9.1 Information on basic physical and chemical properties

Physical state	:	liquid
Color	:	dark, amber
Odor	:	hydrocarbon-like
pH	:	No data available
Melting point/freezing point	:	No data available
Boiling point/boiling range	:	No data available
Flash point	:	55 °C Method: closed cup
Upper explosion limit / Upper flammability limit	:	No data available
Lower explosion limit / Lower flammability limit	:	No data available
Vapor pressure	:	No data available
Relative vapor density	:	No data available
Density	:	7.74 lb/gal
Solubility(ies)		
Water solubility	:	No data available
Solubility in other solvents	:	No data available
Partition coefficient: n-octanol/water	:	No data available
Autoignition temperature	:	No data available
Decomposition temperature	:	No data available
Viscosity		
Viscosity, kinematic	:	No data available

9.2 Other information

Flammability (liquids)	:	Sustains combustion
Particle size	:	Not applicable
Particle Size Distribution	:	Not applicable

SECTION 10: Stability and reactivity

10.1 Reactivity

No decomposition if stored and applied as directed.

10.2 Chemical stability

No decomposition if stored and applied as directed.

10.3 Possibility of hazardous reactions

Hazardous reactions	:	Contact with aqueous acids may produce carbofuran.
---------------------	---	--

MARSHAL® 250 EC

Version	Revision Date:	SDS Number:	Date of last issue: -
1.0	24.09.2025	50000504	Date of first issue: 24.09.2025

Vapors may form explosive mixture with air.

10.4 Conditions to avoid

Conditions to avoid : Heat, flames and sparks.

10.5 Incompatible materials

Materials to avoid : Avoid strong acids, bases, and oxidizers.

10.6 Hazardous decomposition products

Thermal decomposition can lead to release of irritating gases and vapors.

SECTION 11: Toxicological information**11.1 Information on toxicological effects****Acute toxicity**

Toxic if swallowed.
Harmful in contact with skin.
Fatal if inhaled.

Product:

Acute oral toxicity	: LD50 (Rat): 59.9 mg/kg
Acute inhalation toxicity	: LC50 (Rat): 0.265 mg/l Exposure time: 4 h Test atmosphere: dust/mist
Acute dermal toxicity	: LD50 (Rabbit): > 1,520 mg/kg

Components:**Aromatic hydrocarbons, C9; Alkylbenzenes; C9-aromatics:**

Acute oral toxicity	: LD50 (Rat, female): 3,492 mg/kg
Acute inhalation toxicity	: LC50 (Rat, male and female): > 6.193 mg/l Exposure time: 4 h Test atmosphere: vapor Assessment: The substance or mixture has no acute inhalation toxicity
Acute dermal toxicity	: LD50 (Rabbit, male and female): > 3,160 mg/kg

Distillates (petroleum), solvent-dewaxed light paraffinic; Baseoil — unspecified:

Acute oral toxicity	: LD50 (Rat, male and female): > 5,000 mg/kg Method: OECD Test Guideline 401 Remarks: Based on data from similar materials
Acute inhalation toxicity	: LC50 (Rat, male and female): > 5.53 mg/l Exposure time: 4 h Test atmosphere: dust/mist

SAFETY DATA SHEET



MARSHAL® 250 EC

Version 1.0	Revision Date: 24.09.2025	SDS Number: 50000504	Date of last issue: - Date of first issue: 24.09.2025
----------------	------------------------------	-------------------------	--

Assessment: The substance or mixture has no acute inhalation toxicity

Remarks: Based on data from similar materials

Acute dermal toxicity : LD50 (Rabbit, male and female): > 5,000 mg/kg
Method: OECD Test Guideline 402
Remarks: Based on data from similar materials

carbosulfan (ISO):

Acute oral toxicity : LD50 (Rat, female): 185 mg/kg

Acute inhalation toxicity : LC50 (Rat, female): 0.15 mg/l
Exposure time: 4 h
Test atmosphere: dust/mist

Acute dermal toxicity : LD50 (Rat): > 2,000 mg/kg

Sorbitan, tri-(9Z)-9-octadecenoate, poly(oxy-1,2-ethanediyl) derivs.:

Acute oral toxicity : LD50 (Rat): 36,400 mg/kg

Acute toxicity (other routes of administration) : LD50 (Rat): 142 mg/kg

Benzenesulfonic acid, mono-C11-13-branched alkyl derivs., calcium salts:

Acute oral toxicity : LD0 (Rat, male and female): > 2,000 mg/kg
Method: OECD Test Guideline 401
Remarks: no mortality

Acute dermal toxicity : LD50 (Rat, male and female): > 1,000 - 1,600 mg/kg
Method: OECD Test Guideline 402

Skin corrosion/irritation

Repeated exposure may cause skin dryness or cracking.

Product:

Result : slight irritation

Remarks : May cause skin irritation and/or dermatitis.

Components:

Aromatic hydrocarbons, C9; Alkylbenzenes; C9-aromatics:

Species : Rabbit
Method : OECD Test Guideline 404
Result : Mild skin irritation

Assessment : Repeated exposure may cause skin dryness or cracking.

Distillates (petroleum), solvent-dewaxed light paraffinic; Baseoil — unspecified:

Species : Rabbit

SAFETY DATA SHEET



MARSHAL® 250 EC

Version	Revision Date:	SDS Number:	Date of last issue: -
1.0	24.09.2025	50000504	Date of first issue: 24.09.2025

Result : No skin irritation
Remarks : Based on data from similar materials

carbosulfan (ISO):

Species : Rabbit
Result : slight irritation

Sorbitan, tri-(9Z)-9-octadecenoate, poly(oxy-1,2-ethanediyl) derivs.:

Result : No skin irritation

Benzenesulfonic acid, mono-C11-13-branched alkyl derivs., calcium salts:

Species : Rabbit
Result : Skin irritation

Serious eye damage/eye irritation

Causes serious eye irritation.

Product:

Result : Irritation to eyes, reversing within 21 days

Remarks : May cause irreversible eye damage.

Components:

Aromatic hydrocarbons, C9; Alkylbenzenes; C9-aromatics:

Species : Rabbit
Result : No eye irritation

Distillates (petroleum), solvent-dewaxed light paraffinic; Baseoil — unspecified:

Species : Rabbit
Method : OECD Test Guideline 405
Result : No eye irritation
Remarks : Based on data from similar materials

carbosulfan (ISO):

Species : Rabbit
Result : slight irritation

Sorbitan, tri-(9Z)-9-octadecenoate, poly(oxy-1,2-ethanediyl) derivs.:

Species : Rabbit
Result : slight irritation

Benzenesulfonic acid, mono-C11-13-branched alkyl derivs., calcium salts:

Species : Rabbit
Result : Irreversible effects on the eye

MARSHAL® 250 EC

Version	Revision Date:	SDS Number:	Date of last issue: -
1.0	24.09.2025	50000504	Date of first issue: 24.09.2025

Respiratory or skin sensitization**Skin sensitization**

May cause an allergic skin reaction.

Respiratory sensitization

Not classified based on available information.

Product:

Species	:	laboratory animal
Result	:	Probability or evidence of low to moderate skin sensitization rate in humans

Remarks	:	Causes sensitization.
---------	---	-----------------------

Components:**Aromatic hydrocarbons, C9; Alkylbenzenes; C9-aromatics:**

Test Type	:	Maximization Test
Species	:	Guinea pig
Method	:	OECD Test Guideline 406

Distillates (petroleum), solvent-dewaxed light paraffinic; Baseoil — unspecified:

Test Type	:	Buehler Test
Routes of exposure	:	Skin contact
Species	:	Guinea pig
Method	:	OECD Test Guideline 406
Result	:	Not a skin sensitizer.
Remarks	:	Based on data from similar materials

carbosulfan (ISO):

Test Type	:	Buehler Test
Species	:	Guinea pig
Method	:	OECD Test Guideline 406
Result	:	Not a skin sensitizer.

Test Type	:	Patch test
Species	:	Guinea pig
Result	:	May cause sensitization by skin contact.

Sorbitan, tri-(9Z)-9-octadecenoate, poly(oxy-1,2-ethanediyl) derivs.:

Test Type	:	Human repeat insult patch test (HRIPT)
Routes of exposure	:	Skin contact
Species	:	Humans
Result	:	Not a skin sensitizer.

Benzenesulfonic acid, mono-C11-13-branched alkyl derivs., calcium salts:

Test Type	:	Maximization Test
Species	:	Guinea pig
Method	:	OECD Test Guideline 406
Result	:	Does not cause skin sensitization.

MARSHAL® 250 EC

Version	Revision Date:	SDS Number:	Date of last issue: -
1.0	24.09.2025	50000504	Date of first issue: 24.09.2025

Germ cell mutagenicity

Not classified based on available information.

Components:**Aromatic hydrocarbons, C9; Alkylbenzenes; C9-aromatics:**

Genotoxicity in vitro : Test Type: reverse mutation assay
Metabolic activation: with and without metabolic activation
Result: negative

Genotoxicity in vivo : Test Type: Bone marrow chromosome aberration.
Species: Rat
Result: negative

Distillates (petroleum), solvent-dewaxed light paraffinic; Baseoil — unspecified:

Genotoxicity in vitro : Test Type: reverse mutation assay
Metabolic activation: Metabolic activation
Method: OECD Test Guideline 471
Result: positive
Remarks: Based on data from similar materials

Genotoxicity in vivo : Test Type: Micronucleus test
Species: Mouse (male and female)
Application Route: Intraperitoneal injection
Method: OECD Test Guideline 474
Result: negative
Remarks: Based on data from similar materials

carbosulfan (ISO):

Genotoxicity in vitro : Test Type: reverse mutation assay
Test system: Salmonella typhimurium
Result: negative

Test Type: reverse mutation assay
Test system: Escherichia coli
Result: negative

Test Type: gene mutation test
Test system: Chinese hamster cells
Result: negative

Test Type: Chromosome aberration test in vitro
Test system: Chinese hamster cells
Result: negative

Genotoxicity in vivo : Test Type: chromosome aberration assay
Species: mice
Result: negative

Sorbitan, tri-(9Z)-9-octadecenoate, poly(oxy-1,2-ethanediyl) derivs.:

MARSHAL® 250 EC

Version	Revision Date:	SDS Number:	Date of last issue: -
1.0	24.09.2025	50000504	Date of first issue: 24.09.2025

Germ cell mutagenicity- Assessment : Weight of evidence does not support classification as a germ cell mutagen.

Benzenesulfonic acid, mono-C11-13-branched alkyl derivs., calcium salts:

Genotoxicity in vitro : Test Type: In vitro mammalian cell gene mutation test
Result: negative
Remarks: Based on data from similar materials

Test Type: reverse mutation assay
Method: Mutagenicity (Salmonella typhimurium - reverse mutation assay)
Result: negative

Genotoxicity in vivo : Test Type: Micronucleus test
Species: Mouse (male and female)
Application Route: Oral
Result: negative
Remarks: Based on data from similar materials

Germ cell mutagenicity- Assessment : Weight of evidence does not support classification as a germ cell mutagen.

Carcinogenicity

Suspected of causing cancer.

Product:

Carcinogenicity - Assessment : Limited evidence of carcinogenicity in animal studies

Components:

Distillates (petroleum), solvent-dewaxed light paraffinic; Baseoil — unspecified:

Species : Mouse, female
Application Route : Dermal
Exposure time : 78 weeks
Result : negative
Remarks : Based on data from similar materials

carbosulfan (ISO):

Species : Mouse
Exposure time : 2 Years
NOAEL : 2.5 mg/kg bw/day
Result : negative

Species : Rat
Exposure time : 2 Years
NOAEL : 1 mg/kg bw/day
Result : negative

Carcinogenicity - Assessment : Weight of evidence does not support classification as a carcinogen

MARSHAL® 250 EC

Version	Revision Date:	SDS Number:	Date of last issue: -
1.0	24.09.2025	50000504	Date of first issue: 24.09.2025

Sorbitan, tri-(9Z)-9-octadecenoate, poly(oxy-1,2-ethanediyl) derivs.:

Species	: Mouse
Application Route	: Dermal
Result	: negative

Carcinogenicity - Assessment	: Weight of evidence does not support classification as a carcinogen
------------------------------	--

Reproductive toxicity

Not classified based on available information.

Components:**Aromatic hydrocarbons, C9; Alkylbenzenes; C9-aromatics:**

Effects on fertility	: Test Type: Three-generation study Species: Rat Application Route: Inhalation Result: negative
----------------------	--

Effects on fetal development	: Test Type: Pre-natal Species: Rat Application Route: inhalation (vapor) Method: OECD Test Guideline 414 Result: negative Remarks: Based on data from similar materials
------------------------------	---

carbosulfan (ISO):

Effects on fertility	: Test Type: Three-generation study Species: Rat Application Route: Oral General Toxicity Parent: NOAEL: 1.2 mg/kg bw/day Fertility: NOAEL: 1.2 mg/kg bw/day Result: negative
----------------------	--

Effects on fetal development	: Test Type: Embryo-fetal development Species: Rat Application Route: Oral General Toxicity Maternal: NOAEL: 2 mg/kg bw/day Developmental Toxicity: NOAEL: 2 Result: negative
------------------------------	--

	: Test Type: Embryo-fetal development Species: Rabbit Application Route: Oral General Toxicity Maternal: NOAEL: 5 mg/kg bw/day Developmental Toxicity: NOAEL: 10 Result: negative
--	--

Reproductive toxicity - Assessment	: Weight of evidence does not support classification for reproductive toxicity
------------------------------------	--

MARSHAL® 250 EC

Version	Revision Date:	SDS Number:	Date of last issue: -
1.0	24.09.2025	50000504	Date of first issue: 24.09.2025

Sorbitan, tri-(9Z)-9-octadecenoate, poly(oxy-1,2-ethanediyl) derivs.:

Reproductive toxicity - Assessment : Weight of evidence does not support classification for reproductive toxicity

Benzenesulfonic acid, mono-C11-13-branched alkyl derivs., calcium salts:

Effects on fertility : Test Type: Three-generation study
Species: Rat, male and female
Application Route: Oral
Dose: 14, 70, 350 mg/kg bw d
General Toxicity Parent: NOAEL: 350 mg/kg body weight
General Toxicity F1: NOAEL: 350 mg/kg bw/day
General Toxicity F2: NOAEL: 350 mg/kg bw/day
Result: negative
Remarks: Based on data from similar materials

Effects on fetal development : Test Type: reproductive and developmental toxicity study
Species: Rat
Application Route: Oral
Dose: 0.2, 2.0, 300 and 600 mg/kg
Duration of Single Treatment: 20 d
General Toxicity Maternal: LOAEL: 600 mg/kg body weight
Teratogenicity: LOAEL: 600 mg/kg bw/day
Result: negative
Remarks: Based on data from similar materials

Reproductive toxicity - Assessment : Weight of evidence does not support classification for reproductive toxicity

STOT-single exposure

May cause respiratory irritation.
May cause drowsiness or dizziness.
Causes damage to organs.

Product:

Assessment : The substance or mixture is classified as specific target organ toxicant, single exposure, category 1.

Components:**Aromatic hydrocarbons, C9; Alkylbenzenes; C9-aromatics:**

Assessment : May cause respiratory irritation., May cause drowsiness or dizziness.

carbosulfan (ISO):

Target Organs : Nervous system, Bladder, Gastro-intestinal system, Blood
Assessment : The substance or mixture is classified as specific target organ toxicant, single exposure, category 1.

Benzenesulfonic acid, mono-C11-13-branched alkyl derivs., calcium salts:

Assessment : The substance or mixture is not classified as specific target

MARSHAL® 250 EC

Version	Revision Date:	SDS Number:	Date of last issue: -
1.0	24.09.2025	50000504	Date of first issue: 24.09.2025

organ toxicant, single exposure.

STOT-repeated exposure

Causes damage to organs through prolonged or repeated exposure.

Product:

Assessment : The substance or mixture is classified as specific target organ toxicant, repeated exposure, category 1.

Components:**carbosulfan (ISO):**

Target Organs : Nervous system, Bladder, Gastro-intestinal system, Blood
Assessment : The substance or mixture is classified as specific target organ toxicant, repeated exposure, category 1.

Repeated dose toxicity**Components:****Aromatic hydrocarbons, C9; Alkylbenzenes; C9-aromatics:**

Species : Rat, males
NOAEC : 1.8 mg/l
Application Route : inhalation (vapor)
Exposure time : 12 months
Remarks : Based on data from similar materials

carbosulfan (ISO):

Species : Rat
NOAEL : 2 mg/kg bw/day
Application Route : Oral
Exposure time : 90 days

Species : Dog
NOAEL : 1.6 mg/kg bw/day
Application Route : Oral
Exposure time : 6 months

Benzenesulfonic acid, mono-C11-13-branched alkyl derivs., calcium salts:

Species : Rat, male and female
NOAEL : 40 mg/kg bw/day
LOAEL : 115 mg/kg bw/day
Application Route : Oral - feed
Exposure time : 6 months
Dose : 40, 115, 340, 1030 mg/kg bw d
Remarks : Based on data from similar materials

Aspiration toxicity

May be fatal if swallowed and enters airways.

SAFETY DATA SHEET



MARSHAL® 250 EC

Version	Revision Date:	SDS Number:	Date of last issue: -
1.0	24.09.2025	50000504	Date of first issue: 24.09.2025

Product:

The substance or mixture is known to cause human aspiration toxicity hazards or has to be regarded as if it causes a human aspiration toxicity hazard.

Components:

Aromatic hydrocarbons, C9; Alkylbenzenes; C9-aromatics:

May be fatal if swallowed and enters airways.

Distillates (petroleum), solvent-dewaxed light paraffinic; Baseoil — unspecified:

May be fatal if swallowed and enters airways.

carbosulfan (ISO):

The substance does not have properties associated with aspiration hazard potential.

Neurological effects

Components:

carbosulfan (ISO):

Remarks : Neurotoxicity observed in animals studies

Further information

Product:

Remarks : This substance is a reversible cholinesterase-inhibiting pesticide, which elicits symptoms in humans typical of cholinesterase inhibition including headache, light-headedness, weakness, abdominal cramps, nausea, excessive salivation, perspiration and blurred vision. More severe signs of cholinesterase inhibition include tearing, pin-point pupils, excessive respiratory secretions, cyanosis, convulsions, generalized tremor and coma. Excessive cholinesterase inhibition may result in death.

Remarks : Symptoms of overexposure may be headache, dizziness, tiredness, nausea and vomiting.
Concentrations substantially above the TLV value may cause narcotic effects.
Solvents may degrease the skin.

SECTION 12: Ecological information

12.1 Toxicity

Components:

Aromatic hydrocarbons, C9; Alkylbenzenes; C9-aromatics:

Toxicity to fish : LL50 (Oncorhynchus mykiss (rainbow trout)): 9.2 mg/l

SAFETY DATA SHEET



MARSHAL® 250 EC

Version 1.0	Revision Date: 24.09.2025	SDS Number: 50000504	Date of last issue: - Date of first issue: 24.09.2025
----------------	------------------------------	-------------------------	--

Exposure time: 96 h
Method: OECD Test Guideline 203
Remarks: water accommodated fractions (WAF)

Toxicity to daphnia and other aquatic invertebrates : EL50 (Daphnia magna (Water flea)): 3.2 mg/l
Exposure time: 48 h
Method: OECD Test Guideline 202

Toxicity to algae/aquatic plants : NOELR (Pseudokirchneriella subcapitata (green algae)): 0.22 mg/l
Exposure time: 72 h
Method: OECD Test Guideline 201

EL50 (Pseudokirchneriella subcapitata (green algae)): 7.9 mg/l
Exposure time: 72 h
Method: OECD Test Guideline 201

Toxicity to microorganisms : EC50 (activated sludge): > 99 mg/l
Exposure time: 10 min
Method: OECD Test Guideline 209

Ecotoxicology Assessment

Acute aquatic toxicity : Toxic to aquatic life.

Chronic aquatic toxicity : Toxic to aquatic life with long lasting effects.

Distillates (petroleum), solvent-dewaxed light paraffinic; Baseoil — unspecified:

Toxicity to fish : LL50 (Pimephales promelas (fathead minnow)): > 100 mg/l
Exposure time: 96 h
Test Type: static test
Method: OECD Test Guideline 203

Toxicity to daphnia and other aquatic invertebrates : EL50 (Daphnia magna (Water flea)): > 10,000 mg/l
Exposure time: 24 h
Test Type: static test
Method: OECD Test Guideline 202

Toxicity to algae/aquatic plants : NOELR (Pseudokirchneriella subcapitata (green algae)): 100 mg/l
Exposure time: 72 h
Test Type: static test
Method: OECD Test Guideline 201

Toxicity to microorganisms : NOEL : > 1.93 mg/l
Exposure time: 0.16 h

Toxicity to fish (Chronic toxicity) : NOELR: 1,000 mg/l
Exposure time: 14 d
Species: Oncorhynchus mykiss (rainbow trout)

Toxicity to daphnia and other : 10 mg/l

SAFETY DATA SHEET



MARSHAL® 250 EC

Version 1.0	Revision Date: 24.09.2025	SDS Number: 50000504	Date of last issue: - Date of first issue: 24.09.2025
----------------	------------------------------	-------------------------	--

aquatic invertebrates (Chronic toxicity)

Exposure time: 21 d
Species: Daphnia magna (Water flea)
Test Type: semi-static test
Method: OECD Test Guideline 211

carbosulfan (ISO):

Toxicity to fish : LC50 (Lepomis macrochirus (Bluegill sunfish)): 0.015 mg/l
Exposure time: 96 h

Toxicity to daphnia and other aquatic invertebrates : EC50 (Daphnia magna (Water flea)): 0.0015 mg/l
Exposure time: 48 h

Toxicity to algae/aquatic plants : EC50 (Pseudokirchneriella subcapitata (microalgae)): > 20 mg/l
Exposure time: 96 h

M-Factor (Acute aquatic toxicity) : 100

Toxicity to fish (Chronic toxicity) : NOEC: 0.00828 mg/l
Exposure time: 21 d
Species: Pimephales promelas (fathead minnow)

Toxicity to daphnia and other aquatic invertebrates (Chronic toxicity) : NOEC: 0.0032 mg/l
Exposure time: 21 d
Species: Daphnia magna (Water flea)

M-Factor (Chronic aquatic toxicity) : 10

Toxicity to terrestrial organisms : 1.035 µg/bee
Species: Apis mellifera (bees)
Remarks: Oral

0.18 µg/bee
Species: Apis mellifera (bees)
Remarks: Contact

LD50: 10 mg/kg
Species: Anas platyrhynchos (Mallard duck)

Sorbitan, tri-(9Z)-9-octadecenoate, poly(oxy-1,2-ethanediyl) derivs.:

Toxicity to fish : LC50 (Pimephales promelas (fathead minnow)): > 750 mg/l
Exposure time: 96 h
Test Type: static test

Ecotoxicology Assessment

Acute aquatic toxicity : Harmful to aquatic life.

Chronic aquatic toxicity : Harmful to aquatic life with long lasting effects.

Version	Revision Date:	SDS Number:	Date of last issue: -
1.0	24.09.2025	50000504	Date of first issue: 24.09.2025

Benzenesulfonic acid, mono-C11-13-branched alkyl derivs., calcium salts:

- Toxicity to fish : LC50 (Danio rerio (zebra fish)): 31.6 mg/l
Exposure time: 96 h
Method: OECD Test Guideline 203
- Toxicity to daphnia and other aquatic invertebrates : EC50 (Daphnia magna (Water flea)): 62 mg/l
Exposure time: 48 h
Method: OECD Test Guideline 202
- Toxicity to algae/aquatic plants : EC50 (Pseudokirchneriella subcapitata (green algae)): 29 mg/l
Exposure time: 96 h
Remarks: Based on data from similar materials
- NOEC (Pseudokirchneriella subcapitata (green algae)): 0.5 mg/l
Exposure time: 96 h
Remarks: Based on data from similar materials
- Toxicity to microorganisms : EC50 (activated sludge): 550 mg/l
Exposure time: 3 h
Method: OECD Test Guideline 209
- Toxicity to fish (Chronic toxicity) : NOEC: 0.23 mg/l
Exposure time: 72 d
Species: Oncorhynchus mykiss (rainbow trout)
Test Type: flow-through test
Remarks: Based on data from similar materials
- Toxicity to daphnia and other aquatic invertebrates (Chronic toxicity) : NOEC: 1.18 mg/l
Exposure time: 21 d
Species: Daphnia magna (Water flea)
Test Type: flow-through test
Remarks: Based on data from similar materials
- Toxicity to soil dwelling organisms : NOEC: 250 mg/kg
Exposure time: 14 d
Species: Eisenia fetida (earthworms)
Method: OECD Test Guideline 207
Remarks: Based on data from similar materials
- LC50: > 1,000 mg/kg
Exposure time: 14 d
Species: Eisenia fetida (earthworms)
Method: OECD Test Guideline 207
Remarks: Based on data from similar materials
- Plant toxicity : EC50: 167 mg/kg
Exposure time: 21 d
Species: Sorghum bicolor (sorghum)
- 80 mg/kg
Exposure time: 14 d
Species: Avena sativa (oats)

MARSHAL® 250 EC

Version	Revision Date:	SDS Number:	Date of last issue: -
1.0	24.09.2025	50000504	Date of first issue: 24.09.2025

Toxicity to terrestrial organisms : EC10: 82 mg/kg
Exposure time: 21 d
Species: Hypoaspis aculeifer
Remarks: Information given is based on data obtained from similar substances.

12.2 Persistence and degradability**Product:**

Biodegradability : Remarks: No data is available on the product itself.

Components:**Aromatic hydrocarbons, C9; Alkylbenzenes; C9-aromatics:**

Biodegradability : Result: Readily biodegradable.
Biodegradation: 78 %
Exposure time: 28 d
Method: OECD Test Guideline 301F

Distillates (petroleum), solvent-dewaxed light paraffinic; Baseoil — unspecified:

Biodegradability : Result: Inherently biodegradable.
Biodegradation: 31 %
Exposure time: 28 d
Method: OECD Test Guideline 301F

carbosulfan (ISO):

Biodegradability : Result: Not readily biodegradable.
Biodegradation: 28 %
Exposure time: 28 d

Stability in water : Remarks: Hydrolyzes readily.

Sorbitan, tri-(9Z)-9-octadecenoate, poly(oxy-1,2-ethanediyl) derivs.:

Biodegradability : Result: Not readily biodegradable.
Remarks: Based on data from similar materials

Benzenesulfonic acid, mono-C11-13-branched alkyl derivs., calcium salts:

Biodegradability : Inoculum: activated sludge, non-adapted
Result: Not readily biodegradable.
Biodegradation: 2.9 %
Exposure time: 28 d
Method: OECD Test Guideline 301E

Result: Inherently biodegradable.
Biodegradation: > 35 - 45 %
Exposure time: 10 d

SAFETY DATA SHEET

MARSHAL® 250 EC



Version	Revision Date:	SDS Number:	Date of last issue: -
1.0	24.09.2025	50000504	Date of first issue: 24.09.2025

12.3 Bioaccumulative potential

Product:

Bioaccumulation : Remarks: No data is available on the product itself.

Components:

Aromatic hydrocarbons, C9; Alkylbenzenes; C9-aromatics:

Partition coefficient: n-octanol/water : log Pow: 2.92 - 3.59
Method: QSAR

carbosulfan (ISO):

Bioaccumulation : Species: Fish
Bioconcentration factor (BCF): 990
Remarks: Can accumulate in aquatic organisms.

Partition coefficient: n-octanol/water : log Pow: 5.37
pH: 8
Method: OECD Test Guideline 107

Sorbitan, tri-(9Z)-9-octadecenoate, poly(oxy-1,2-ethanediyl) derivs.:

Bioaccumulation : Remarks: Bioaccumulation is unlikely.
Based on data from similar materials

Benzenesulfonic acid, mono-C11-13-branched alkyl derivs., calcium salts:

Bioaccumulation : Bioconcentration factor (BCF): 3.16
Method: QSAR

Partition coefficient: n-octanol/water : log Pow: 4.595 (20 °C)

12.4 Mobility in soil

Product:

Distribution among environmental compartments : Remarks: No data is available on the product itself.

Components:

carbosulfan (ISO):

Distribution among environmental compartments : Remarks: Slightly mobile in soils

Stability in soil :

12.5 Results of PBT and vPvB assessment

Product:

Assessment : This substance/mixture contains no components considered

SAFETY DATA SHEET



MARSHAL® 250 EC

Version	Revision Date:	SDS Number:	Date of last issue: -
1.0	24.09.2025	50000504	Date of first issue: 24.09.2025

to be either persistent, bioaccumulative and toxic (PBT), or very persistent and very bioaccumulative (vPvB) at levels of 0.1% or higher.

12.6 Other adverse effects

Product:

Endocrine disrupting potential : The substance/mixture does not contain components considered to have endocrine disrupting properties according to REACH Article 57(f) or Commission Delegated regulation (EU) 2017/2100 or Commission Regulation (EU) 2018/605 at levels of 0.1% or higher.

Additional ecological information : An environmental hazard cannot be excluded in the event of unprofessional handling or disposal.
Very toxic to aquatic life with long lasting effects.

SECTION 13: Disposal considerations

13.1 Waste treatment methods

Product : The product should not be allowed to enter drains, water courses or the soil.
Do not contaminate ponds, waterways or ditches with chemical or used container.
Send to a licensed waste management company.

Contaminated packaging : Empty remaining contents.
Triple rinse containers.
Do not re-use empty containers.
Packaging that is not properly emptied must be disposed of as the unused product.
Empty containers should be taken to an approved waste handling site for recycling or disposal.

SECTION 14: Transport information

14.1 UN number

UNRTDG : UN 2991
IMDG : UN 2991
IATA : UN 2991

14.2 UN proper shipping name

UNRTDG : CARBAMATE PESTICIDE, LIQUID, TOXIC, FLAMMABLE
(Carbosulfan, Xylene)
IMDG : CARBAMATE PESTICIDE, LIQUID, TOXIC, FLAMMABLE
(Carbosulfan, Xylene)

SAFETY DATA SHEET



MARSHAL® 250 EC

Version	Revision Date:	SDS Number:	Date of last issue: -
1.0	24.09.2025	50000504	Date of first issue: 24.09.2025

IATA : Carbamate pesticide, liquid, toxic, flammable
(Carbosulfan, Xylene)

14.3 Transport hazard class(es)

	Class	Subsidiary risks
UNRTDG	: 6.1	3
IMDG	: 6.1	3
IATA	: 6.1	3

14.4 Packing group

UNRTDG
Packing group : II
Labels : 6.1 (3)

IMDG
Packing group : II
Labels : 6.1 (3)
EmS Code : F-E, S-D

IATA (Cargo)
Packing instruction (cargo aircraft) : 662
Packing instruction (LQ) : Y641
Packing group : II
Labels : Toxic, Flammable Liquids

IATA (Passenger)
Packing instruction (passenger aircraft) : 654
Packing instruction (LQ) : Y641
Packing group : II
Labels : Toxic, Flammable Liquids

14.5 Environmental hazards

UNRTDG
Environmentally hazardous : yes

IMDG
Marine pollutant : yes

IATA (Passenger)
Environmentally hazardous : yes

IATA (Cargo)
Environmentally hazardous : yes

14.6 Special precautions for user

The transport classification(s) provided herein are for informational purposes only, and solely based upon the properties of the unpackaged material as it is described within this Safety Data Sheet. Transportation classifications may vary by mode of transportation, package sizes, and variations in regional or country regulations.

14.7 Transport in bulk according to Annex II of Marpol and the IBC Code

Not applicable for product as supplied.

MARSHAL® 250 EC

Version	Revision Date:	SDS Number:	Date of last issue: -
1.0	24.09.2025	50000504	Date of first issue: 24.09.2025

SECTION 15: Regulatory information**15.1 Safety, health and environmental regulations/legislation specific for the substance or mixture****The ingredients of this product are reported in the following inventories:**

TCSI	: Not in compliance with the inventory
TSCA	: Product contains substance(s) not listed on TSCA inventory.
AIIC	: Not in compliance with the inventory
DSL	: This product contains chemical substance(s) exempt from CEPA DSL Inventory requirements. It is regulated as a pesticide subject to Pest Control Products Act (PCPA) requirements. Read the PCPA label, authorized under the Pest Control Products Act, prior to using or handling this pest control product.
ENCS	: Not in compliance with the inventory
ISHL	: Not in compliance with the inventory
KECI	: Not in compliance with the inventory
PICCS	: Not in compliance with the inventory
IECSC	: Not in compliance with the inventory
NZIoC	: Not in compliance with the inventory
TECI	: Not in compliance with the inventory

15.2 Chemical Safety Assessment

No Chemical Safety Assessment has been carried out for this mixture.

SECTION 16: Other information**Full text of H-Statements**

H226	: Flammable liquid and vapor.
H301	: Toxic if swallowed.
H304	: May be fatal if swallowed and enters airways.
H312	: Harmful in contact with skin.
H315	: Causes skin irritation.
H317	: May cause an allergic skin reaction.
H318	: Causes serious eye damage.
H330	: Fatal if inhaled.

MARSHAL® 250 EC

Version	Revision Date:	SDS Number:	Date of last issue: -
1.0	24.09.2025	50000504	Date of first issue: 24.09.2025

H335	:	May cause respiratory irritation.
H336	:	May cause drowsiness or dizziness.
H350	:	May cause cancer.
H370	:	Causes damage to organs.
H372	:	Causes damage to organs through prolonged or repeated exposure.
H400	:	Very toxic to aquatic life.
H410	:	Very toxic to aquatic life with long lasting effects.
H411	:	Toxic to aquatic life with long lasting effects.
H412	:	Harmful to aquatic life with long lasting effects.

Full text of other abbreviations

Acute Tox.	:	Acute toxicity
Aquatic Acute	:	Short-term (acute) aquatic hazard
Aquatic Chronic	:	Long-term (chronic) aquatic hazard
Asp. Tox.	:	Aspiration hazard
Carc.	:	Carcinogenicity
Eye Dam.	:	Serious eye damage
Flam. Liq.	:	Flammable liquids
Skin Irrit.	:	Skin irritation
Skin Sens.	:	Skin sensitization
STOT RE	:	Specific target organ toxicity - repeated exposure
STOT SE	:	Specific target organ toxicity - single exposure

ADN - European Agreement concerning the International Carriage of Dangerous Goods by Inland Waterways; ADR - Agreement concerning the International Carriage of Dangerous Goods by Road; AIIIC - Australian Inventory of Industrial Chemicals; ASTM - American Society for the Testing of Materials; bw - Body weight; CLP - Classification Labelling Packaging Regulation; Regulation (EC) No 1272/2008; CMR - Carcinogen, Mutagen or Reproductive Toxicant; DIN - Standard of the German Institute for Standardisation; DSL - Domestic Substances List (Canada); ECHA - European Chemicals Agency; EC-Number - European Community number; ECx - Concentration associated with x% response; ELx - Loading rate associated with x% response; EmS - Emergency Schedule; ENCS - Existing and New Chemical Substances (Japan); ErCx - Concentration associated with x% growth rate response; GHS - Globally Harmonized System; GLP - Good Laboratory Practice; IARC - International Agency for Research on Cancer; IATA - International Air Transport Association; IBC - International Code for the Construction and Equipment of Ships carrying Dangerous Chemicals in Bulk; IC50 - Half maximal inhibitory concentration; ICAO - International Civil Aviation Organization; IECSC - Inventory of Existing Chemical Substances in China; IMDG - International Maritime Dangerous Goods; IMO - International Maritime Organization; ISHL - Industrial Safety and Health Law (Japan); ISO - International Organisation for Standardization; KECI - Korea Existing Chemicals Inventory; LC50 - Lethal Concentration to 50 % of a test population; LD50 - Lethal Dose to 50% of a test population (Median Lethal Dose); MARPOL - International Convention for the Prevention of Pollution from Ships; n.o.s. - Not Otherwise Specified; NO(A)EC - No Observed (Adverse) Effect Concentration; NO(A)EL - No Observed (Adverse) Effect Level; NOELR - No Observable Effect Loading Rate; NZIoC - New Zealand Inventory of Chemicals; OECD - Organization for Economic Co-operation and Development; OPPTS - Office of Chemical Safety and Pollution Prevention; PBT - Persistent, Bioaccumulative and Toxic substance; PICCS - Philippines Inventory of Chemicals and Chemical Substances; (Q)SAR - (Quantitative) Structure Activity Relationship; REACH - Regulation (EC) No 1907/2006 of the European Parliament and of the Council concerning the Registration, Evaluation, Authorisation and Restriction of Chemicals; RID - Regulations concerning the International Carriage of Dangerous Goods by Rail; SADT - Self-Accelerating Decomposition Temperature; SDS - Safety Data Sheet; SVHC - substance of very high concern; TCSI - Taiwan Chemical Substance Inventory; TECI - Thailand Existing Chemicals Inventory; TRGS - Technical Rule for Hazardous Substances; TSCA

SAFETY DATA SHEET



MARSHAL® 250 EC

Version	Revision Date:	SDS Number:	Date of last issue: -
1.0	24.09.2025	50000504	Date of first issue: 24.09.2025

- Toxic Substances Control Act (United States); UN - United Nations; vPvB - Very Persistent and Very Bioaccumulative

Further information

Other information :

Classification of the mixture:

Flam. Liq. 3	H226
Acute Tox. 3	H301
Acute Tox. 2	H330
Acute Tox. 4	H312
Eye Irrit. 2	H319
Skin Sens. 1B	H317
Carc. 2	H351
STOT SE 1	H370
STOT SE 3	H335
STOT SE 3	H336
STOT RE 1	H372
Asp. Tox. 1	H304
Aquatic Acute 1	H400
Aquatic Chronic 1	H410

Classification procedure:

Based on product data or assessment
Based on product data or assessment
Based on product data or assessment
Based on product data or assessment
Based on product data or assessment
Based on product data or assessment
Based on product data or assessment
Based on product data or assessment
Calculation method
Calculation method
Based on product data or assessment
Based on product data or assessment
Calculation method
Calculation method

Disclaimer

FMC Corporation believes that the information and recommendations contained herein (including data and statements) are accurate as of the date hereof. You can contact FMC Corporation to insure that this document is the most current available from FMC Corporation. No warranty of fitness for any particular purpose, warranty of merchantability or any other warranty, expressed or implied, is made concerning the information provided herein. The information provided herein relates only to the specified product designated and may not be applicable where such product is used in combination with any other materials or in any process. The user is responsible for determining whether the product is fit for a particular purpose and suitable for the user's conditions and methods of use. Since the conditions and methods of use are beyond the control of FMC Corporation, FMC Corporation expressly disclaims any and all liability as to any results obtained or arising from any use of the products or reliance on such information.

Prepared by

FMC Corporation

FMC and the FMC Logo are trademarks of FMC Corporation and/or an affiliate.

© 2021-2025 FMC Corporation. All Rights Reserved.

KE / EN

SAFETY DATA SHEET



MARSHAL® 250 EC

Version	Revision Date:	SDS Number:	Date of last issue: -
1.0	24.09.2025	50000504	Date of first issue: 24.09.2025
